

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456157.8234 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.06
Transit depth (Rp/Rs)^2	0.95 +/- 1.17 [%]
Orbital Inclination (inc)	87.86 +/- 1.9
Airmass coefficient 1 (a1)	0.855 +/- 0.094
Airmass coefficient 2 (a2)	0.15 +/- 0.47
Scatter in the residuals of the lightcurve fit is	10.95 %
Best Comparison Star	#1 - [309, 141]
Optimal Aperture	1.97
Optimal Annulus	11.26
Transit Duration (day)	0.074 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.097 ± 0.06 vs 0.1592 (deviation 1.04σ)
Duration fit vs published	0.074 d vs 0.0946 d (deviation 0.94σ)
Mid-time O–C	-3.6 min
Depth SNR	1.6
Residual scatter	10.95 %

Light curve

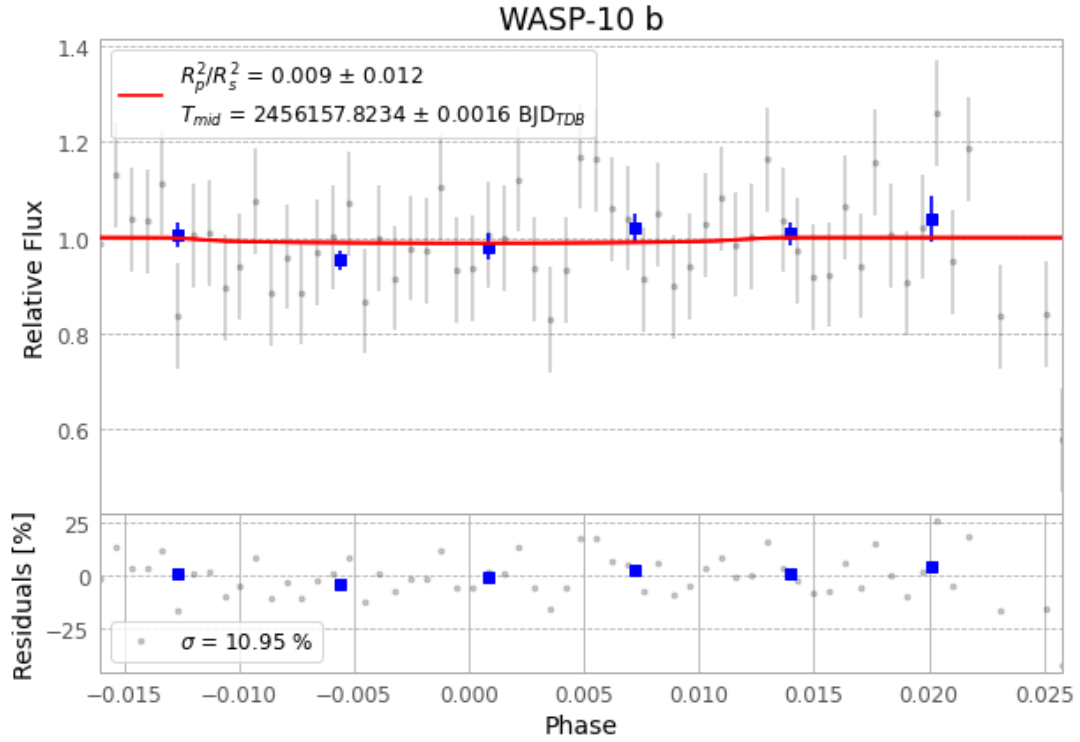


Figure 1 — FinalLightCurve_WASP-10 b_2012-08-17.png (SHA-256 47324ba5d122162e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5be79a661aace2c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), WASP-10120818062712.FITS → WASP-10120818094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-120818.log (09609e37a205...), FinalLightCurve_WASP-10 b_2012-08-17.png (47324ba5d122...), FinalLightCurve_WASP-10 b_2012-08-17.csv (d1a006395a62...)

Compute resources

Wall time 292.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:11Z.